

Defining Probability

- In everyday conversation, probability is a measure of someone's belief in the occurrence of a future event
- Quantitatively, defined as the fraction of times an event occurs if it's repeated many times (Relative frequency)
- Axiom of probability (More rigorous)

Probability vs Statistics

- Probability uses the randomization mechanism is assumed, and calculations follow
 - Ex. calculating odd for 8/10 heads on a fair coin
- Statistics uses the outcome to infer something about the mechanism
 - Ex. asking whether a coin is fair given 8/10 heads

Uncertainty

- Randomness/Uncertainty because the particular outcomes are uncontrollable/unpredictable